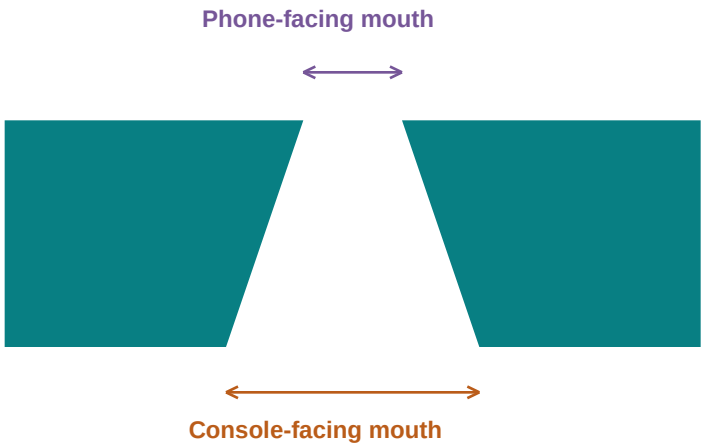
First, measure the failed print

The old 5.6 / 13.5 mm opening targets are no longer fixed requirements. Measure the actual plastic before cleaning up or sanding away evidence; remove loose supports first.

Record the first contact. Try each insert separately, without the phone and without forcing it. Note whether it stops on a nub side, lower end, upper end, or the console corner. A tight nub can hold the whole part up and make the corner look wrong.



Use calipers lightly.
Keep the inside jaw tips at the mouth being measured. Do not let them drop into the wider or narrower section.

Measure straight-slot width near the lower end, at the middle, and near the upper end, outside the rounded end caps.

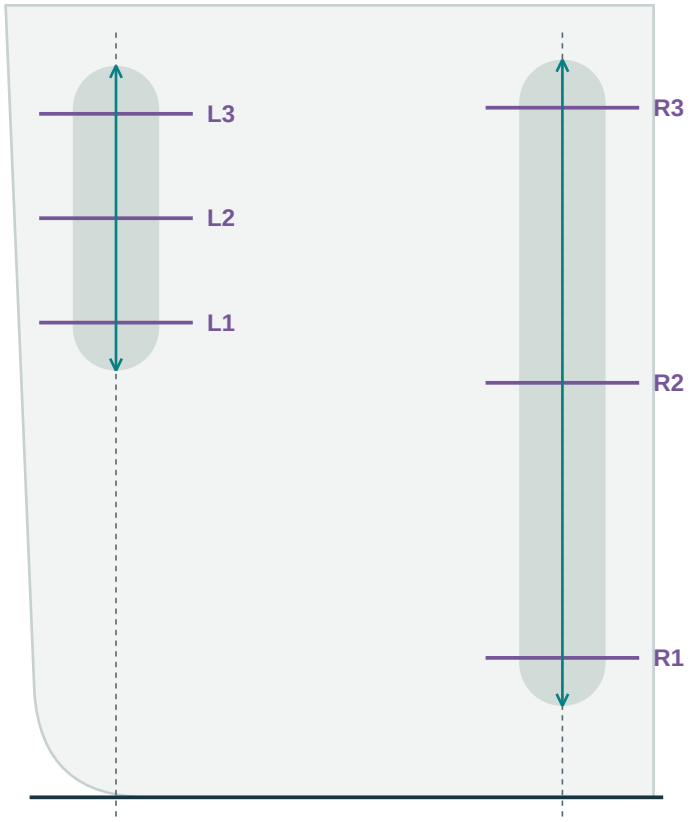
V6 opening	CAD reference only	Your actual printed measurement
Left, phone-facing	Width 5.6; total length 60.9	Widths lower / middle / upper: ___ / ___ / ___ Total length: ___
Left, console-facing	Width 13.5; total length 64.9	Widths lower / middle / upper: ___ / ___ / ___ Total length: ___
Right, phone-facing	Width 5.6; total length 125.02	Widths lower / middle / upper: ___ / ___ / ___ Total length: ___
Right, console-facing	Width 13.5; total length 136.02	Widths lower / middle / upper: ___ / ___ / ___ Total length: ___

Also record: which ends rub; any rough support residue; and the gap under the lower-left corner and the far end when the part rests freely. Use a feeler gauge or a paper stack whose thickness you measure. Take one photo showing the caliper contact and reading.

For widths, use stations about 10 mm inside each finished end and the midpoint; record the actual station if different. Measure total length in the pad-plane direction. Report inaccessible readings as such. Photos of the actual contact are more useful than a forced measurement.

Capture the real nub sections

Remove both inserts. Six cross-sections, plus one centerline length profile per nub, give the depth-dependent shape. The sloped print meets each nub at a different height.



Factory lower crease / Y = 0

Datums
Y = 0: inside factory lower crease, extended through the corner.
+Y: toward the camera end, along the pad.
X = 0: short left nub centerline; +X to the right.
Z = 0: adjacent flat pad. Z points perpendicular to the pad, not vertically in the car.

At every section
Capture both flanks, the crown and a flat pad baseline. Hold the gauge face in the section plane and its pins normal to the pad. Use light pressure; do not compress the fabric.

Positions drawn here use the old model only. **Record your actual Y values.** Do not assume the old model coordinates are correct.

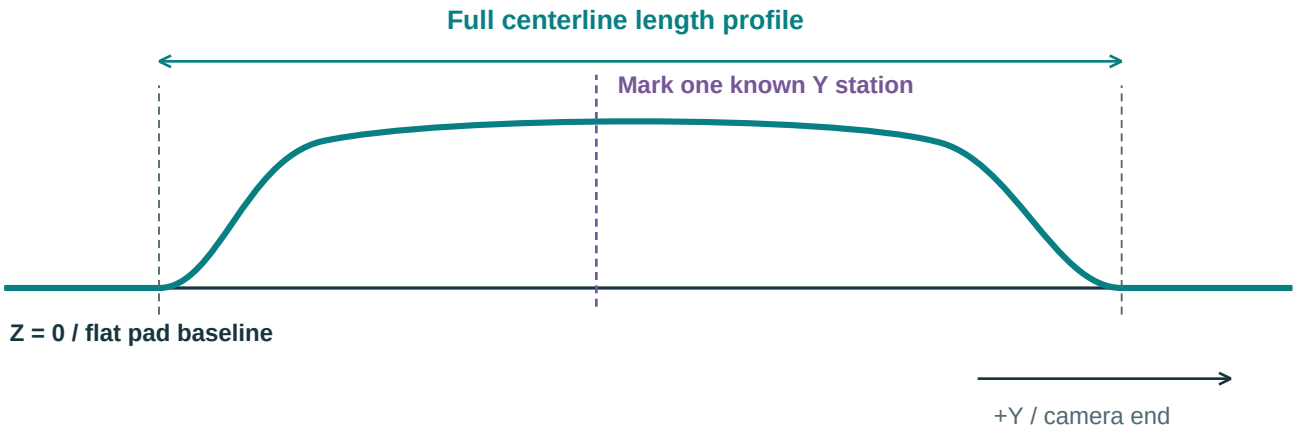
ID	Position on the real nub	Actual Y from crease	Max. height / base width*
L1 / R1	10 mm above its lower end	L ___ / R ___	L ___ / ___; R ___ / ___
L2 / R2	Midpoint of its actual length	L ___ / R ___	L ___ / ___; R ___ / ___
L3 / R3	10 mm below its upper end	L ___ / R ___	L ___ / ___; R ___ / ___

*Use calipers or a depth bridge for independent checks where accessible. Record the reference plane and contact height; do not claim a base width if the fabric blend makes its boundary unclear. A traced full section is the main input.

I will extract the widths at each local print height. V6 used Z = 1 + 0.02123Y: for example, Y=30 meets Z=1.64, while Y=120 meets Z=3.55. Measuring every nub only at 4.4 mm would miss the lower interference.

Locate the ends and trace the shape

A curve alone is not enough: its flat-pad baseline, direction and one known position must travel with it. Paper grid coordinates are not console coordinates.



Two length profiles: LL and RL. Run the gauge along each nub centerline and include flat pad beyond both ends. If the long nub exceeds the gauge, use two captures with at least 25 mm of shared, marked overlap. Put a known Y station on each trace; do not align the pieces by eye.

Bare-console measurement	Left nub	Right nub
Lower footprint end, Y from crease	___ mm	___ mm
Upper footprint end, Y from crease	___ mm	___ mm
Total footprint length, independently checked	___ mm	___ mm
Centerline profile ID and known Y mark	LL: Y ___ at marked pin	RL: Y ___ at marked pin

One alignment check: at a recorded Y where both nubs exist, measure their center-to-center spacing using the midpoint of each pair of flanks at the same height. Record Y ___, contact Z ___, spacing ___. The old value was 93 mm; do not force the new data to match it.

Keep the actual size. Lock the gauge, mark the pin matching the known station, and trace the contact-tip edge onto flat grid paper with a fine pencil held upright. Mark Z=0, +Y (or +X), the station and which nub. Photograph the tracing straight down with grid visible around it. Take a separate placement photo.

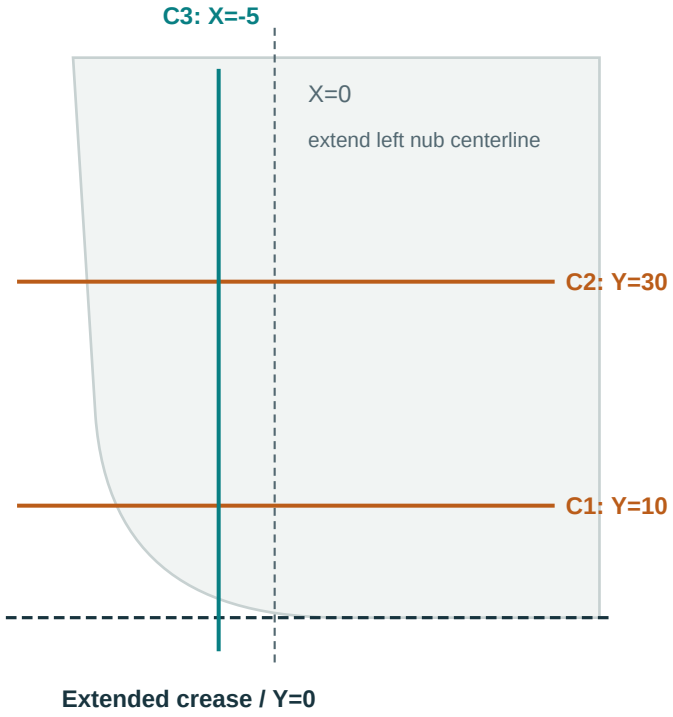
End outline, if accessible: include a straight-on overhead photo of each bare nub with a ruler close to the pad, especially all four rounded tips. This helps distinguish the end shape; the registered length profiles control dimensions. Do not infer an exact radius from the photograph.

Measure the corner in two directions

First establish the nub interference. Do not treat the corner lift of a nub-bound insert as the amount to remove from the corner.

C1 and C2: upright sections across the left-wall-to-pad transition, 10 and 30 mm above the extended crease. Include flat pad.

C3: upright section along the pad, 5 mm left of the short nub centerline. Capture the base transition and continue upward until a flat pad baseline is visible.



Record actual stations if these lines cannot be reached. X=-5 is a section location, not a proposed edge position.

All three traces must share the same pad and crease references.

Corner record	What to provide
C1 / C2 / C3	Actual Y or X; locked-gauge tracing with baseline and direction; placement photo.
CP0: low outline	Trim thin paper/card until it lies flat in the lower-left corner without curling. Capture at least 30 mm along each adjoining edge. Record material thickness and any visible gaps.
CP1: 1 mm outline	If practical, repeat with the contact edge held parallel to the pad at Z=1.0 mm. Measure total contact height including the paper/card; use equal supports. If this cannot be controlled, mark it unavailable.

On both corner templates, mark the extended Y=0 crease, the extended X=0 nub centerline (or a measured offset to it), and +Y. Photograph the template in place, then flat against the grid. **Do not bend paper into the corner and then flatten it:** that does not preserve a plan outline.

These records distinguish the plan outline from the vertical wall/base blend. A single guessed radius or diagonal profile cannot determine both. The existing R15 corner is provisional and may change.

Return this set, then test locally

The next full assembly should follow successful nub and corner fit tests. CAD validity alone did not establish physical fit in V6.

Priority	Return items	What it resolves
1	Printed slot width readings and four total lengths from page 1; first-contact notes; installed photos.	Print undersize/support residue versus an incorrectly sized CAD opening.
2	L1-L3 and R1-R3: six located transverse traces. LL and RL: two full longitudinal profiles, split with overlap if needed.	Real width and end recession at each height, including the sloped upper rim.
3	Four bare-nub end Y positions; total lengths; one center-spacing check where accessible.	Absolute position and length, independent of rounded edge treatments.
4	C1-C3: three upright corner traces; CP0 flat template; CP1 if its height can be controlled.	Lower-left corner plan fit and the underside wall/base transition separately.

Every trace needs: ID, actual station, flat-pad baseline, direction, one known positional mark where needed, a placement photo and a flat tracing photo. Keep the ruler/grid in the tracing plane. The horizontal 100 mm grid bar is already confirmed; check one vertical 100 mm span too.

Repeatability: take each gauge profile twice. Overlay the tracings and record their largest difference. If they differ visibly, take a third and note why. Record the pin pitch and contact-tip thickness if you can. Report actual readings; do not round a rough tracing to 0.1 mm or compress the padded surface to make it repeat.

What I will change in the next design: use the measured curves at their actual scale and locations; size the openings around the real nub envelope plus tested clearance; calculate each section at its local print height; and separate printer compensation from physical clearance. The old 5.6 / 13.5 mm sizes and 6 / 5 mm end reductions will not override a demonstrated interference.

Before another complete holder: prepare separate full-height nub collars with a small set of clearance choices, then an independently located corner test. Include flat-pad seating references and check their positions from the crease. Test without force. Consistent seating and no rocking establish the fit before combining the pieces.

The confirmed phone/camera dimensions and 4.5 mm hard thickness limit remain. The first task is console seating. Lens contact and charging are separate checks after the inserts and phone can seat correctly.